Supplementary Appendices — Invariant Game Representation Learning from Steam Reviews

Supplement to the anonymous submission; the paper is at https://anonymous.4open.science/r/latenarray-I2CE. Appendix letters and table numbers are as cited in the paper.

APPENDIX A: THE I-CE / CE ABLATION — CONTRAST ALONE VERSUS CONTRAST WITH INVARIANCE

This ablation removes the invariance term from I-CE and changes nothing else (Figure 5, Tables A1 and A2).

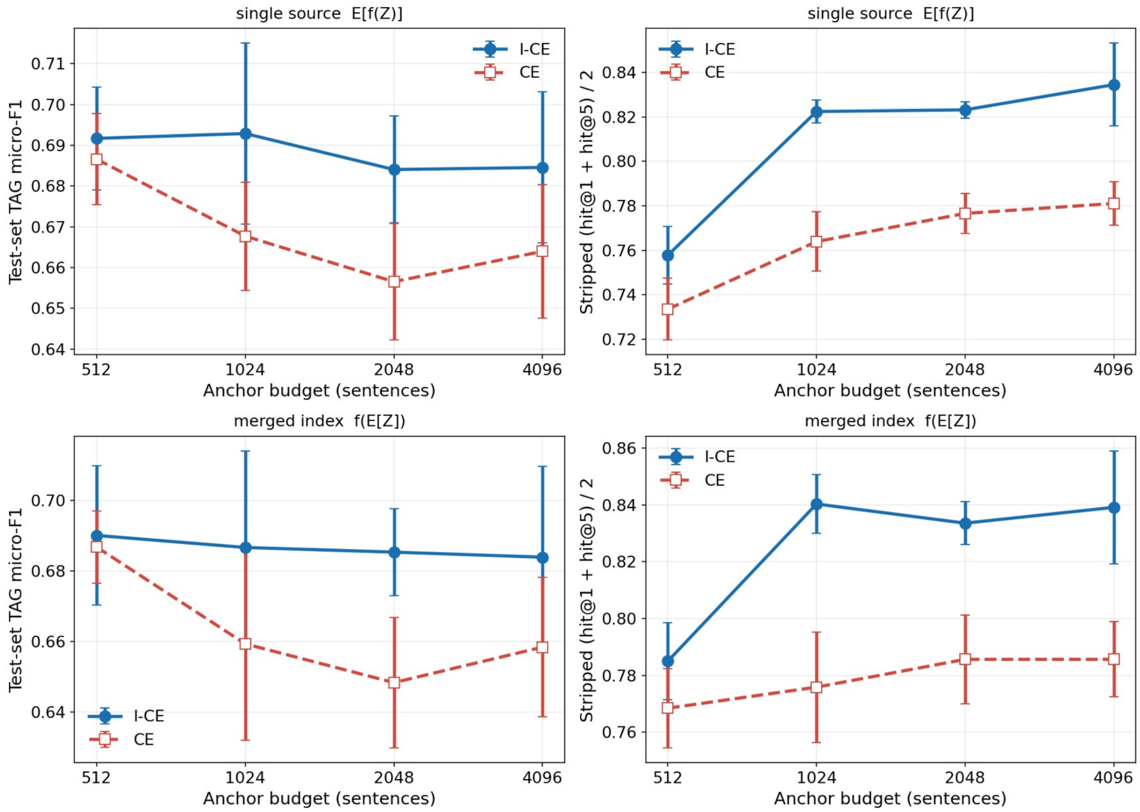

Figure 5: The I-CE / CE ablation across four anchor budgets (five-fold mean ± std; stripped = name-stripped queries; per-budget numbers in Appendix Table A2). Left: test-set TAG micro-F1. Right: stripped (hit@1 + hit@5)/2. Top row: the single-source reading the body uses; bottom row: the merged-index reading of Appendix F. The budget buys one step — 512 to 1,024 sentences — and then a plateau, and the I-CE curve sits above CE at every budget on both axes and in both regimes.

Table A1: Per-fold detail at the 4,096-sentence anchor budget, testset queries (ts). I-CE beats CE on Stripped hit@1 in five of five paired folds at this budget, and in 20 of 20 across the four budgets of Table A2.

Fold	**Name hit@1**	**Name hit@5**	**Stripped hit@1**	**Stripped hit@5**	**Test-set TAG F1**
I-CE fold 0	0.970	1.000	0.728	0.920	0.689
I-CE fold 1	0.955	1.000	0.745	0.931	0.665
I-CE fold 2	0.931	0.989	0.715	0.904	0.662
I-CE fold 3	0.952	0.994	0.749	0.922	0.697
I-CE fold 4	0.962	1.000	0.769	0.963	0.709
I-CE mean ± std	**0.954 ± 0.013**	**0.997 ± 0.004**	**0.741 ± 0.018**	**0.928 ± 0.020**	**0.685 ± 0.019**
CE mean ± std	**0.938 ± 0.013**	**0.988 ± 0.010**	**0.673 ± 0.014**	**0.889 ± 0.011**	**0.664 ± 0.016**

Table A2: The I-CE / CE ablation in full — CE vs I-CE at anchor budgets of 512–4,096 sentences (testset queries (ts), five-fold mean ± std, the two objectives seed-paired per fold on identical splits). The budget buys one step — 512 to 1,024 sentences, the only significant one ($p = 0.003$) — and then a plateau in which the three larger budgets are indistinguishable ($p \geq 0.20$); I-CE leads CE at every budget (20 of 20 paired folds on Stripped hit@1), so the invariance constraint pays at all four scales. The windowed teacher (Appendix D) appears at its trained budget.

Objective	**Name hit@1**	**Name hit@5**	**Stripped hit@1**	**Stripped hit@5**	**Test-set TAG F1**
CE @512	0.891 ± 0.018	0.983 ± 0.005	0.614 ± 0.011	0.853 ± 0.019	0.687 ± 0.011
I-CE @512	**0.916 ± 0.009**	**0.991 ± 0.003**	**0.644 ± 0.024**	**0.872 ± 0.010**	**0.692 ± 0.013**
CE @1024	0.922 ± 0.016	0.988 ± 0.011	0.655 ± 0.007	0.873 ± 0.022	0.668 ± 0.013
I-CE @1024	**0.942 ± 0.015**	**0.997 ± 0.002**	**0.727 ± 0.006**	**0.918 ± 0.006**	**0.693 ± 0.022**
CE @2048	0.929 ± 0.011	0.989 ± 0.007	0.671 ± 0.015	0.882 ± 0.014	0.657 ± 0.014
I-CE @2048	**0.947 ± 0.016**	**0.996 ± 0.003**	**0.728 ± 0.009**	**0.918 ± 0.010**	**0.684 ± 0.013**
CE @4096	0.938 ± 0.013	0.988 ± 0.010	0.673 ± 0.014	0.889 ± 0.011	0.664 ± 0.016
I-CE @4096	**0.954 ± 0.013**	**0.997 ± 0.004**	**0.741 ± 0.018**	**0.928 ± 0.020**	**0.685 ± 0.019**
swin-I-CE @4096 (windowed)	0.934 ± 0.014	0.994 ± 0.004	0.709 ± 0.027	0.922 ± 0.010	0.691 ± 0.014

APPENDIX B: THE BASELINE RECIPES OF TABLE 2

Every trained baseline shares the full scaffolding of Section 4 — the same 4-query cross-attention tower, the same four views per game per step (three review views and one document view), the same corpus, batch (192 games), optimizer (AdamW, lr 5e-4, weight decay 1e-4), 2,000-epoch budget, temperature $\tau = 0.02$ where a softmax exists, and the validation-only selection of Section 5. They differ only in the loss.

Frozen embedder (mean pool). No tower and no training: a game is the L2-normalized mean of its anchor-pack sentence embeddings, and a query is the L2-normalized mean of its own sentences.

CE (contrast only). Equation (2) alone: each of the four strong views is classified against the full anchor gallery, re-encoded with gradient at every step; the invariance term of Equation (1) is removed.

SimCLR-style (in-batch views). No anchors anywhere in training. The step's 4 × 192 view encodings form the key set of an NT-Xent loss [4]: view v of a game takes its ring sibling — view v+1 (mod 4) of the same game — as the positive, the game's other own views are masked as false negatives, and every other game's views serve as negatives at $\tau = 0.02$. The tower is unchanged; only the contrast lives in view space instead of anchor space. The anchor gallery is encoded once at evaluation time, by the finished tower.

VICReg (epd 20/10/20). No negatives and no gallery. Each view encoding passes through an expander MLP, and the VICReg loss [1] is applied to expander outputs over the six view pairs with variance/invariance/covariance weights 20/10/20; views are drawn for 192 games per step, with the variance term estimating per-dimension spread within the step's view batch. Anchors are encoded only at evaluation time.

BYOL. An online tower plus a two-layer predictor head, and a target tower that is an exponential moving average of the online weights (momentum 0.996) behind a stop-gradient. The loss is 1 − cos between the predictor's output for one view and the target tower's encoding of another, averaged over ordered view pairs; the document view participates like any other. Collapse is prevented by the predictor/EMA asymmetry rather than by negatives.

CE + anchor-in-I (Table 4). Equation (2) unchanged, but the invariance term of Equation (1) ranges over the ten pairs among the four strong views and the game's own anchor, instead of the six pairs among views alone. It is the weak-view arm of Section 6.2: the same constant-weight pull on the anchor that the decomposed objectives use, with the softmax left in place. Two-stage: BYOL warm start → windowed (Table 5). Stage one is the BYOL recipe above, run to its own selection point; stage two re-initializes the tower from those weights and trains the windowed teacher of Appendix D for 600 epochs, 30% of the from-scratch discriminative budget. Selection happens inside stage two. I-CE (EMA + memory bank; Table 5). The fully coupled teacher is replaced by an EMA shadow tower (momentum 0.99) that encodes the current batch's anchors into a 3,072-key memory bank; each view runs a cross-entropy against the ring (its freshly written key is the positive, stale same-game keys are masked), and the invariance term is unchanged. Gradient reaches the strong-view side only.

APPENDIX C: THE DOCUMENT VIEW — PRESENT, ABSENT, REWRITTEN

The document view is one of the four strong views, and this ablation isolates it at the 512-sentence anchor budget (Table A3). Wikipedia-derived text is exactly the material a public sentence embedder is likely to have memorized; a faithful sentence-wise rewrite preserves the content while breaking the verbatim match, so it acts as a decontaminating filter.

Table A3: The document view — present, absent, rewritten (testset queries (ts), five-fold mean ± std at the 512-sentence anchor budget, validation-selected checkpoints, zero-shot, seed-paired folds). The document view is worth 0.008 Stripped hit@1 (and 0.032 Name hit@1), and rewriting it helps: the LLM-rewritten row leads raw documents on the stripped columns (+0.008 Stripped hit@1) and ties on the name columns. The rewrite is not a cost but a gain.

Objective	Name hit@1	Name hit@5	Stripped hit@1	Stripped hit@5	Test-set TAG F1
I-CE, document view present	**0.916 ± 0.009**	**0.991 ± 0.003**	**0.644 ± 0.024**	**0.872 ± 0.010**	**0.692 ± 0.013**
I-CE, no document view	0.883 ± 0.016	0.981 ± 0.007	0.636 ± 0.016	0.871 ± 0.013	0.680 ± 0.024
I-CE, LLM-rewritten documents	0.914 ± 0.010	0.990 ± 0.002	0.652 ± 0.023	0.882 ± 0.009	0.689 ± 0.016

APPENDIX D: THE WINDOWED TEACHER AND THE WINDOW-SIZE, VICREG, AND TEMPERATURE SWEEPS

swin-I-CE bounds the fully coupled teacher's memory ceiling. The catalog is laid out on a ring with a persistent pointer p. Each step runs two micro-passes: micro-pass l ∈ {0, 1} freshly encodes the $W = 168$ anchors starting at ring position p + l·S (stride $S = 84$, so consecutive passes overlap by half a window) together with the current batch's own anchors, forms its own CE partition — every view's positive sits in the batch block, the window anchors serve as negatives — and backpropagates immediately, freeing the window's activations before the next pass; the pointer then advances by 2S, sweeping the full catalog every ten steps so that every anchor takes a regular, gradient-coupled turn. The tower, the views, and the invariance term are unchanged.

Table A4: The window-size sweep behind the windowed teacher (fixed split, testset queries (ts) of that split, 4,096-sentence anchors, validation-selected checkpoints; a single anchor draw rather than the ten-draw expectation of the main tables). Coverage counts the training-fold anchor packs re-encoded with gradient per step (batch plus the union of the two micro-pass windows). Halving or doubling the window moves nothing beyond single-split noise — the smallest window already holds the plateau, so memory, not coverage, should pick W. Tag readings are the test-set probe of Section 5, recomputed on this split so that they sit on the same scale as the rest of the paper; their spread here is single-split noise, and at fold level swin and full coupling read tags at parity (Table 5, which gives the five-fold account of $W = 168$).

Objective	Name hit@1	Name hit@5	Stripped hit@1	Stripped hit@5	Test-set TAG F1
swin, $W = 84$ (step 42, 19.7% coverage)	0.946	1.000	0.691	0.902	0.714
swin, $W = 168$ (step 84, 27.5% coverage)	**0.941**	**0.995**	**0.691**	**0.892**	**0.704**
swin, $W = 336$ (step 168, 43.1% coverage)	0.941	1.000	0.686	0.897	0.668
I-CE, fully coupled (100% coverage)	**0.951**	**1.000**	**0.706**	**0.917**	**0.679**

Table A5: What the sliding window bounds. The anchor gallery is the activation ceiling of the fully coupled objective — it re-encodes every training anchor pack with gradient at once — whereas swin holds only the batch's own anchors plus one $W = 168$ window per micro-pass and frees it before the next, so the gradient-coupled encoding at its peak is 360 anchors, 22% of the full graph. Both variants keep the whole catalog resident: swin bounds the activation, not the store, which is why measured peak GPU memory falls by roughly 40% at the 4,096-sentence budget rather than the full 4.5× — the balance is the persistent ring. Anchor packs are capped at 4,096 sentences and realized packs are shorter, so the sentence-slot counts are ceilings. Counts are quoted on the fixed split of Appendix D (1,613 training games); a five-fold gallery is 1,694 and scales the same way.

	Full-gallery I-CE	**swin-I-CE ($W = 168$)**
Anchors re-encoded with gradient per backward (peak)	1,613	360 (192 batch + 168 window)
as a fraction of the training gallery	100%	22%
Sentence-slots in the backward graph (4,096 budget)	≈ 6.6 M	≈ 1.5 M
Anchor packs held resident (the ring)	1,613 (all)	1,613 (all)

The VICReg grid below is older than the rest of the protocol: it predates validation selection and its towers were logged only at trajectory peaks, so Table A6 reports each arm's peak stripped hit@1 and omits tag readings, which cannot be re-scored under the test-set protocol of Section 5.

Table A6: The VICReg weight sweep (fixed split, testset queries (ts) of that split, 512-sentence anchors at evaluation, a single anchor draw; trajectory peaks). All rows use the canonical wiring — variance, invariance, and covariance all on the expander-output pair; the earlier centroid wiring (invariance as an MSE between unit-norm view centroids) collapsed retrieval outright at every view width (hit@1 ≤ 0.03) and is omitted. V/I/C = 20/10/20 is the best cell and the image-domain 25/25/1 trails it. Drawing views for the whole training pool per step (batch = all) buys true population moments in the variance term at roughly eight times the step cost; its one completed fold scored within fold noise of the batch-192 recipe (stripped hit@1 0.387 under the single-draw protocol, against that recipe's 0.440 ± 0.021 five-fold mean under the averaged-anchor protocol), so Table 2's five-fold row trains at batch 192. I-CE and CE peaks under the same readout anchor the scale.

Objective	**Name hit@1**	**Stripped hit@1**
V/I/C = 25/25/1	0.373	0.216
V/I/C = 20/10/20 (recipe weights)	**0.578**	**0.284**
V/I/C = 20/10/15	0.574	0.265
25/25/1, batch = all	0.328	0.201
20/10/20, batch = all	0.495	0.250
I-CE (reference, same readout)	**0.926**	**0.672**
CE (reference, same readout)	0.922	0.618

Table A7: The temperature sweep behind the Section 7 verdict (testset queries (ts), five-fold mean ± std at the 512- and 2,048-sentence anchor budgets; the $\tau = 0.10$ cell at 512 has four folds). $\tau = 0.02$ is retrieval-optimal at both budgets: softening to 0.05/0.10 pays 0.060–0.226 stripped hit@1, and a learnable temperature drifts to the identity-first end (+0.010 stripped, +0.008 tag). What softening buys instead is the tag reading — $\tau = 0.05$ lifts it into the negative-light band of Section 6.1 — so temperature is a tag/identity dial, not an untapped optimum.

Objective	Name hit@1	Name hit@5	Stripped hit@1	Stripped hit@5	Test-set TAG F1
$\tau = 0.02$ (recipe), 512	**0.916 ± 0.009**	**0.991 ± 0.003**	**0.644 ± 0.024**	**0.872 ± 0.010**	**0.692 ± 0.013**
$\tau = 0.05$, 512	0.896 ± 0.017	0.987 ± 0.007	0.584 ± 0.021	0.859 ± 0.014	0.704 ± 0.017
$\tau = 0.10$, 512 (four folds)	0.828 ± 0.027	0.956 ± 0.020	0.490 ± 0.019	0.770 ± 0.016	0.701 ± 0.021
$\tau = 0.02$ (recipe), 2,048	**0.947 ± 0.016**	**0.996 ± 0.003**	**0.728 ± 0.009**	**0.918 ± 0.010**	**0.684 ± 0.013**
$\tau = 0.05$, 2,048	0.938 ± 0.018	0.992 ± 0.004	0.655 ± 0.027	0.890 ± 0.015	0.710 ± 0.015
$\tau = 0.10$, 2,048	0.883 ± 0.032	0.978 ± 0.014	0.502 ± 0.012	0.785 ± 0.012	0.707 ± 0.015
learnable τ, 2,048	0.951 ± 0.008	0.995 ± 0.005	0.738 ± 0.006	0.920 ± 0.015	0.693 ± 0.015

APPENDIX E: THE WEAK VIEW MUST EXCLUDE THE STRONG ONES

The fixed anchor pack never contains the sentences a strong view just read. In arm vfa, at every step, each batch game's anchor pack is opened with the game's four fresh strong views and the fixed pack fills the remainder, so the weak view becomes a superset of the strong ones; evaluation anchors stay the fixed packs.

Table A8: Views-first anchors (vfa), a softer weak view that fails. Each batch game's anchor pack is opened with the game's own four strong views before the fixed pack; five-fold @512 (testset queries (ts), a single anchor draw per fold) against the identical-split I-CE reference. Retrieval collapses on both axes (zero of five paired folds) and the tag reading does not rise; every fold selects the epoch-50 checkpoint. The TAG column is the review-probe reading (the @512 references predate the held-out test-set probe), consistent within this table. Both rows also predate the selection rule of Section 5 and share the earlier criterion, so the comparison is internal to the table; the collapse it reports is not a selection artifact, since every fold of the softer weak view picks its earliest checkpoint and degrades from there.

Objective	Name hit@1	Stripped hit@1	TAG F1
Views-first anchors (vfa) @512	0.638 ± 0.055	0.398 ± 0.041	0.670 ± 0.021
I-CE (reference, same split) @512	0.901 ± 0.021	0.657 ± 0.028	0.705 ± 0.015

APPENDIX F: THE TWO ANCHOR-READING REGIMES

Table A9: The single-source reading — each of the ten anchor packs is scored on its own and the scores are averaged (five-fold mean ± std, testset queries (ts), 4,096-sentence anchors). This is the protocol of every table in the body, reproduced here for comparison with Table A10.

Objective	Name hit@1	Name hit@5	Stripped hit@1	Stripped hit@5	Test-set TAG F1
I-CE (ours)	0.954 ± 0.013	0.997 ± 0.004	0.741 ± 0.018	0.928 ± 0.020	0.685 ± 0.019
swin-I-CE (~27% gradient window)	0.934 ± 0.014	0.994 ± 0.004	0.709 ± 0.027	0.922 ± 0.010	0.691 ± 0.014
Two-stage: BYOL warm start → windowed, 600 ep	0.930 ± 0.011	0.991 ± 0.008	0.709 ± 0.019	0.917 ± 0.005	0.705 ± 0.022
CE (contrast only)	0.938 ± 0.013	0.988 ± 0.010	0.673 ± 0.014	0.889 ± 0.011	0.664 ± 0.016
SimCLR-style (in-batch views)	0.918 ± 0.024	0.990 ± 0.007	0.632 ± 0.029	0.882 ± 0.022	0.707 ± 0.019
I-CE (EMA + memory bank)	0.936 ± 0.010	0.991 ± 0.007	0.636 ± 0.016	0.881 ± 0.016	0.711 ± 0.016
VICReg (expander wiring, v/i/c = 20/10/20)	0.823 ± 0.010	0.960 ± 0.016	0.451 ± 0.017	0.729 ± 0.033	0.708 ± 0.022
BYOL	0.441 ± 0.033	0.737 ± 0.026	0.284 ± 0.017	0.553 ± 0.043	0.712 ± 0.017
Frozen embedder (mean pool)	0.425 ± 0.033	0.626 ± 0.014	0.209 ± 0.020	0.380 ± 0.031	0.588 ± 0.009

Table A10: The merged-index reading — the ten anchor encodings are averaged into one index vector per game before scoring (same towers, folds and testset queries (ts) as Table A9). Pooling adds at most 0.008 Stripped hit@1 and no method changes rank.

Objective	Name hit@1	Name hit@5	Stripped hit@1	Stripped hit@5	Test-set TAG F1
I-CE (ours)	0.956 ± 0.019	0.996 ± 0.005	0.747 ± 0.021	0.931 ± 0.020	0.684 ± 0.026
swin-I-CE (~27% gradient window)	0.941 ± 0.018	0.994 ± 0.004	0.716 ± 0.026	0.926 ± 0.011	0.681 ± 0.019
Two-stage: BYOL warm start → windowed, 600 ep	0.932 ± 0.014	0.991 ± 0.008	0.716 ± 0.019	0.920 ± 0.005	0.706 ± 0.020
CE (contrast only)	0.950 ± 0.012	0.989 ± 0.008	0.678 ± 0.019	0.893 ± 0.013	0.658 ± 0.020
SimCLR-style (in-batch views)	0.919 ± 0.027	0.990 ± 0.006	0.634 ± 0.027	0.886 ± 0.021	0.704 ± 0.021

I-CE (EMA + memory bank)	0.944 ± 0.013	0.993 ± 0.006	0.641 ± 0.012	0.887 ± 0.013	0.717 ± 0.014
VICReg (expander wiring, v/i/c = 20/10/20)	0.830 ± 0.012	0.962 ± 0.020	0.453 ± 0.014	0.735 ± 0.037	0.705 ± 0.025
BYOL	0.447 ± 0.034	0.743 ± 0.022	0.284 ± 0.014	0.554 ± 0.046	0.715 ± 0.020
Frozen embedder (mean pool)	0.450 ± 0.034	0.642 ± 0.015	0.215 ± 0.031	0.393 ± 0.035	0.578 ± 0.015

APPENDIX G: TRAINSET VERSUS TESTSET QUERIES

Every headline number in this paper is measured on testset queries (ts) — games held out of the fold the tower trained on. Because the 814-game evaluation universe also contains games inside the training pool, the same measurement can be repeated on trainset queries (tr): about 488 per fold against 163 testset queries, ranked against the same 2,020 candidates by the same finished tower.

Table A11: Retrieval on trainset queries (tr) versus testset queries (ts), five-fold means at the 4,096-sentence anchor budget, single-source reading. Gap = (tr) − (ts). The frozen row is the control: no training, no gap.

Objective	Name hit@1 (tr)	Name hit@1 (ts)	Gap	Stripped hit@1 (tr)	Stripped hit@1 (ts)	Gap
I-CE (ours)	0.991	0.954	+0.037	0.740	0.741	-0.002
swin-I-CE (~27% gradient window)	0.983	0.934	+0.049	0.716	0.709	+0.007
Two-stage: BYOL warm start → windowed, 600 ep	0.979	0.930	+0.049	0.708	0.709	-0.000
CE (contrast only)	0.986	0.938	+0.048	0.735	0.673	+0.062
SimCLR-style (in-batch views)	0.987	0.918	+0.069	0.655	0.632	+0.023
I-CE (EMA + memory bank)	0.978	0.936	+0.042	0.613	0.636	-0.024
VICReg (expander wiring, v/i/c = 20/10/20)	0.902	0.823	+0.079	0.480	0.451	+0.029
BYOL	0.732	0.441	+0.291	0.355	0.284	+0.072
Frozen embedder (mean pool)	0.425	0.425	+0.000	0.209	0.209	-0.000

APPENDIX H: COST AT SCALE

Write N for the catalog size, P for the anchor-pack cap in sentences, $d = 1{,}024$ for the embedding width, $B = 192$ for the games in a step, and $W = 168$, $S = 84$, $L = 2$ for the window, stride and micro-passes of the teacher in Appendix D. The tower reads one pack with $Q = 4$ latent queries, so encoding a pack costs $\Theta(P \cdot Q \cdot d)$ work and holds $\Theta(P \cdot d)$ activation.

The fully coupled teacher re-encodes every pack with gradient at every step. Its cost is $\Theta(N \cdot P \cdot Q \cdot d)$ in time and $\Theta(N \cdot P \cdot d)$ in memory — both linear in the catalog. At our scale the backward graph holds 6.9 M sentence slots, 13 GiB of fp16 input before any activation is stored (Table A5).

The windowed teacher removes N from both expressions. A micro-pass encodes the batch's own packs plus one window and frees them before the next, so a step costs $\Theta(L \cdot (B + W) \cdot P \cdot Q \cdot d)$ in time and $\Theta((B + W) \cdot P \cdot d)$ in memory. At $B = 192$ and $W = 168$ that is 1.47 M sentence slots, 2.8 GiB of input, whether the catalog holds two thousand entries or ten million. What the window does not bound is the ring itself: the windowed teacher still keeps every anchor pack resident (Table A5), so $\Theta(N \cdot P \cdot d)$ of store sits on the device even though only $\Theta((B + W) \cdot P \cdot d)$ of it is ever differentiated.

Table A12: The cost model of the three variants, with the fully coupled I-CE teacher normalized to 1 at the same catalog size. N is the catalog, P the anchor-pack cap, d the embedding width, $Q = 4$ latent queries, $B = 192$ games per step, and $W = 168$, $S = 84$, $L = 2$ the window, stride and micro-passes. Ratios are quoted at our $N = 1{,}694$; the windowed teacher's step contains no N, so both of its first two rows fall as 1/N, but its ring keeps every pack resident and therefore stays linear in the catalog on the device. The streamed store — the full-corpus pipeline of this paper — is what removes that last term, paying I/O for it. Measured figures are at $P = 4{,}096$; the last row solves each variant's device budget for N on one 80 GiB accelerator.

Quantity	I-CE, fully coupled	swin-I-CE ($W = 168$)	swin + streamed store
Step time	$\Theta(N \cdot P \cdot Q \cdot d)$ **– 1**	$\Theta(L(B+W) \cdot P \cdot Q \cdot d$ **– 0.43**	$\Theta(L(B+W) \cdot P \cdot$ **– 0.43**
Peak activation	$\Theta(N \cdot P \cdot d)$ **– 1**	$\Theta((B+W) \cdot P \cdot d)$ **– 0.21**	$\Theta((B+W) \cdot P \cdot d$ **– 0.21**
Resident anchor store	$\Theta(N \cdot P \cdot d)$ **– 1**	$\Theta(N \cdot P \cdot d)$ **– 1**	$\Theta((B+W) \cdot P \cdot d$ **– 0.21**
Device memory in N	**linear**	**linear**	**constant**
Per-step device I/O	**none**	**none**	$\Theta(L \cdot S \cdot P \cdot d)$ **≈ 1.3 GiB**
Measured at $N = 1{,}694$	**≈ 45 GiB**	**≈ 27 GiB**	**≈ 14 GiB**
Largest N on one 80 GiB device	**≈ 3,000**	**≈ 8,500**	**not memory-bound**

The third variant closes that last term by giving up residency. The merged pool is a single memory-mapped fp16 array on disk: the build writes and reads it a million rows at a time, so that assembling 23,373 games never holds more than a chunk, and the training loop gathers each anchor pack from that map instead of from a resident ring. The device then holds only what it differentiates, $\Theta\left(\left(B + W\right) \cdot P \cdot d\right)$, and the catalog leaves device memory altogether. What comes back is I/O: the ring advances L·$S = 168$ packs per step and only the new ones must be paged, $\Theta\left(L \cdot S \cdot P \cdot d\right)$ — 1.3 GiB per step at $P = 4{,}096$, 336 MiB at $P = 1{,}024$ — which a prefetch thread hides behind the forward pass. The 23,373-game corpus needs 183 GiB of packs at $P = 4{,}096$, and 46 GiB at $P = 1{,}024$. An anchor takes a gradient turn once every N/(L·S) steps — ten steps at our $N = 1{,}694$, 139 at 23,373, and 5,952, about 370 epochs of sixteen steps, at a million; equivalently, per-step coverage is (B + W)/N, falling from 21% here to 1.5% at full-corpus scale and 0.04% at a million. A million-entity store is 7.6 TiB at $P = 4{,}096$ or 1.9 TiB at $P = 1{,}024$.

APPENDIX I: THE QUERY-REWRITE PROMPTS

All four query registers come from one chat-completions model (temperature 0.7) that reads a held-out game's full wiki article and writes an English description of it. The instructions below were issued in Chinese and are translated faithfully here; the verbatim strings, the retry policy and the model identifier are in the released code. An output shorter than 300 characters is re-requested with the elaboration instruction, up to three attempts, and the article is truncated to 12,000 characters before it is sent.

ALGORITHM 1: The rewrite instructions behind the four query registers

user message — "Game: {title}" then the article text

neutral (Name hit@k) — Read the full game page text the user provides and summarize it into a descriptive article about the game. Preserve the real information about the game's content, gameplay and mechanics in full; do not invent anything that is not on the page. Register: neutral. Write in English and output the article body directly.

name-stripped (Stripped hit@k) — Read the full game page text the user provides and summarize it into a descriptive article about the game, in a neutral register. Preserve the real information about the game's content, gameplay and mechanics in full; do not invent anything that is not on the page. But do not reveal the game's name, the characters' names, the items' names or anything like them — replace them all with imagined, invented words. Write in English and output the article body directly.

praising (Appendix K) — the neutral instruction with the register clause replaced by: Register: praising and affirming.

critical (Appendix K) — the neutral instruction with the register clause replaced by: Register: negative and critical.

elaboration retry (any register) — Your previous answer was too short. Please write a considerably MORE DETAILED article (at least 300 words) in the requested style, covering the game's content, story, world and mechanics from the page text above.

APPENDIX J: REDUCING THE STEAM TAG VOCABULARY

A single mapping file is the source of truth for every tag number in this paper: each of the 202 fine tags is assigned either to one coarse class or to the sentinel "del".

The listing below is the whole definition: a class is present on a game if any of the Steam tags on its right-hand side is, and the vote weights Steam attaches are used only to resolve which source is strongest, never as a target — the probe predicts presence, so a game either carries a class or does not. On our 2,020 games this yields a 2,020 × 23 binary matrix of density 0.202 — 4.6 classes per game at the mean, 5 at the median, and between 35 and 1,395 games per class.

ALGORITHM 2: The 23 TAG classes, each written as the original Steam tags it absorbs

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Action/Adventure = [Action, Action-Adventure, Adventure, Combat, Hack and Slash, Souls-like]
Bullet Hell = [Bullet Hell]
Card Game = [Card Game, Card Battler, Deckbuilding, Roguelike Deckbuilder]
Co-op/Multiplayer = [Class-Based, Co-op, Co-op Campaign, Competitive, Local Co-Op, Local Multiplayer, Massively
Multiplayer, Multiplayer, Online Co-Op, PvE, PvP, Split Screen, Team-Based]
Crime/Mystery = [Assassin, Crime, Detective, Mystery, Political]
Fantasy = [Fantasy, Dark Fantasy, Dragons, Gothic, Magic, Medieval, Mythology]
Fighting/Melee = [3D Fighter, Beat 'em up, Character Action Game, Fighting, Martial Arts, Spectacle fighter,
Swordplay]
Historical/War = [Alternate History, Historical, Military, War, World War II]
Horror = [Horror, Demons, Psychological Horror, Supernatural, Survival Horror, Vampire, Zombies]
Narrative/Choices = [Choices Matter, Choose Your Own Adventure, Conversation, Interactive Fiction, Lore-Rich,
Multiple Endings, Narration, Narrative, Point & Click, Story Rich, Text-Based, Visual Novel, Walking Simulator]
Open World/Survival = [Base-Building, Building, Crafting, Exploration, Fishing, Hunting, Mining, Open World, Open
World Survival Craft, Sandbox, Survival]
Platformer = [Platformer, 2D Platformer, 3D Platformer, Metroidvania, Parkour]
Puzzle = [Puzzle]
RPG = [RPG, Action RPG, CRPG, JRPG, MMORPG, Party-Based RPG, Strategy RPG, Tactical RPG]
Racing/Driving = [Driving, Racing, Tanks, Vehicular Combat]
Roguelike = [Action Roguelike, Dungeon Crawler, Procedural Generation, Rogue-like, Rogue-lite]
Sci-fi/Cyberpunk = [Aliens, Cyberpunk, Dystopian, Futuristic, Post-apocalyptic, Robots, Sci-fi, Space]
Shooter/FPS = [FPS, Looter Shooter, Shooter, Third-Person Shooter]
Simulation/Management = [Agriculture, Automation, Automobile Sim, City Builder, Colony Sim, Farming Sim, Life
Sim, Management, Resource Management, Simulation, Space Sim]
Sports/Rhythm = [Rhythm, Sports]
Stealth/Immersive = [Immersive Sim, Stealth]
Strategy/Tactics = [4X, Grand Strategy, RTS, Real Time Tactics, Strategy, Tactical, Tower Defense, Wargame]
Turn-Based = [Turn-Based, Turn-Based Combat, Turn-Based Strategy, Turn-Based Tactics]

discarded = [2D, 3D, Anime, Arcade, Atmospheric, Beautiful, Cartoony, Casual, Character Customization, Cinematic,
Classic, Colorful, Comedy, Controller, Cute, Dark, Dark Humor, Dating Sim, Destruction, Difficult, Drama, Early
Access, Economy, Emotional, Family Friendly, Fast-Paced, Female Protagonist, First-Person, Free to Play, Funny, Gore,
Great Soundtrack, Gun Customization, Hand-drawn, Indie, Inventory Management, Isometric, LGBTQ+, Loot, Mature,
Memes, Music, Nudity, Old School, Physics, Pixel Graphics, Psychedelic, Psychological, Realistic, Relaxing, Replay
Value, Retro, Romance, Sexual Content, Singleplayer, Soundtrack, Stylized, Surreal, Third Person, Thriller, Violent]
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APPENDIX K: TONE-REWRITTEN QUERIES

Table A13: Tone-rewritten queries (testset queries (ts), hit@1 among all 2,020 games, five-fold mean ± std; same towers, checkpoints and protocol as Table 2). Each held-out game's article is rewritten by the same model in a praising and in a critical voice (Appendix I); names stay intact, so the Neutral column is Table 2's Name hit@1, and the last column is the paired per-fold difference. Tone moves retrieval by at most 0.05 anywhere — against the 0.2+ that removing names costs (Table 2). The two anchored-contrast towers are the only ones a critical voice never helps — I-CE loses in five folds of five, CE in four — while the four other spaces gain on average, the frozen embedder most (+0.051).

Objective	**Positive hit@1**	**Neutral hit@1**	**Negative hit@1**	**Negative − Neutral**
Frozen embedder	0.435 ± 0.015	0.425 ± 0.033	0.476 ± 0.025	+0.051 ± 0.032
BYOL	0.494 ± 0.022	0.441 ± 0.033	0.475 ± 0.020	+0.034 ± 0.030
VICReg (epd 20/10/20)	0.849 ± 0.027	0.823 ± 0.010	0.833 ± 0.021	+0.010 ± 0.028
SimCLR-style (in-batch views)	0.924 ± 0.018	0.918 ± 0.024	0.926 ± 0.015	+0.008 ± 0.031
CE (contrast only)	0.933 ± 0.018	0.938 ± 0.013	0.927 ± 0.019	−0.011 ± 0.011
I-CE (ours)	0.957 ± 0.011	0.954 ± 0.013	0.943 ± 0.021	−0.011 ± 0.009

Table A14: TAG micro-F1 on the same tone-rewritten queries (five-fold mean ± std; the probes are Table 2's TAG column exactly — one ridge per gallery draw, K = 10, held-out protocol of Section 5 — so the Neutral column is Table 2's TAG column and only the query register changes). The direction inverts against Table A13: every trained tower reads tags from the critical rewrite as well as or better than from the neutral one — I-CE +0.012 in five folds of five — while the frozen embedder loses −0.018 in five of five. The identity cost a critical voice carries in Table A13 does not extend to the semantic reading.

Objective	**Positive TAG F1**	**Neutral TAG F1**	**Negative TAG F1**	**Negative − Neutral**
Frozen embedder	0.591 ± 0.005	0.588 ± 0.009	0.570 ± 0.008	−0.018 ± 0.004
BYOL	0.719 ± 0.014	0.712 ± 0.017	0.719 ± 0.015	+0.007 ± 0.004
VICReg (epd 20/10/20)	0.714 ± 0.017	0.708 ± 0.022	0.713 ± 0.017	+0.004 ± 0.006
SimCLR-style (in-batch views)	0.711 ± 0.016	0.707 ± 0.019	0.713 ± 0.015	+0.007 ± 0.005
CE (contrast only)	0.676 ± 0.016	0.664 ± 0.016	0.672 ± 0.009	+0.008 ± 0.010
I-CE (ours)	0.699 ± 0.016	0.685 ± 0.019	0.697 ± 0.014	+0.012 ± 0.008